ABS INDICATOR (Continued)

position and within about two seconds check for continuity between the ABS warning indicator driver circuit cavity of the instrument panel wire harness connector (Connector C2) for the instrument cluster and a good ground. There should be continuity for about two seconds after ignition On, and then an open circuit. If OK, proceed to the diagnosis for the ABS brake system. (Refer to 5 - BRAKES - DIAGNOSIS AND TESTING). If not OK, go to Step 2.

(2) Turn the ignition switch to the Off position. Disconnect and isolate the battery negative cable. Remove the Controller Antilock Brake (CAB) relay from the Power Distribution Center (PDC). Check for continuity between the ABS warning indicator driver circuit cavity of the instrument panel wire harness connector (Connector C2) for the instrument cluster and a good ground. There should be no continuity. If OK, go to Step 3. If not OK, repair the shorted ABS warning indicator driver circuit between the instrument cluster and the PDC as required.

(3) Check for continuity between the ABS warning indicator driver circuit cavities of the instrument panel wire harness connector (Connector C2) for the instrument cluster and the CAB relay receptacle in the PDC. There should be continuity. If OK, proceed to the diagnosis for the ABS brake system. (Refer to 5 - BRAKES - DIAGNOSIS AND TESTING). If not OK, repair the open ABS warning indicator driver circuit between the instrument cluster and the PDC as required.

AIRBAG INDICATOR

DESCRIPTION

Fig. 8 Airbag Indicator

An airbag indicator is standard equipment on all instrument clusters (Fig. 8). However, the instrument cluster can be programmed to disable this indicator on vehicles that are not equipped with the airbag system, which is not available in some markets. The airbag indicator is located near the lower edge of the instrument cluster, to the right of center. The airbag indicator consists of a stencil-like cutout of the International Control and Display Symbol icon for “Airbag” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A red Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the icon to appear in red through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The airbag indicator is serviced as a unit with the instrument cluster.

OPERATION

The airbag indicator gives an indication to the vehicle operator when the airbag system is faulty or inoperative. The airbag indicator is controlled by a transistor on the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Airbag Control Module (ACM) over the Programmable Communications Interface (PCI) data bus. The airbag indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a battery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the airbag indicator for the following reasons:

- **Bulb Test** - Each time the ignition switch is turned to the On position the airbag indicator is illuminated for about seven seconds. The first two seconds is the cluster bulb test function, and the remainder is the ACM bulb test function.
- **ACM Lamp-On Message** - Each time the cluster receives a lamp-on message from the ACM, the airbag indicator will be illuminated. The indicator remains illuminated for about twelve seconds or until the cluster receives a lamp-off message from the ACM, whichever is longer.
- **Communication Error** - If the cluster receives no airbag messages for five consecutive seconds, the airbag indicator is illuminated. The indicator remains illuminated until the cluster receives a single lamp-off message from the ACM.
- **Actuator Test** - Each time the cluster is put through the actuator test, the airbag indicator will be turned on, then off again during the bulb check portion of the test to confirm the functionality of the LED and the cluster control circuitry. The actuator test illumination of the airbag indicator is a function of the ACM.

The ACM continually monitors the airbag system circuits and sensors to decide whether the system is in good operating condition. The ACM then sends the proper lamp-on or lamp-off messages to the instrument cluster. If the ACM sends a lamp-on message after the bulb test, it indicates that the ACM has detected a system malfunction and/or that the airbags may not deploy when required, or may deploy